

W 3

HTTP 通訊協定與網頁資料請求

請求解剖 · 查詢參數與 URL 編碼 · 開發者工具 Network

吳彥霖 博士

Python 爬蟲實務應用 · 第 3 週 · 09/29

三節課，三件事

節 1

HTTP 請求/回應解剖

- URL 結構
- headers 與 User-Agent
- logging 執行紀錄

節 2

requests 參數

- params 與 URL 編碼
- 用 resp.url 驗證

節 3

開發者工具 Network

- 看瀏覽器真正送出的請求

節 1

HTTP 請求 / 回應解剖

爬蟲機制

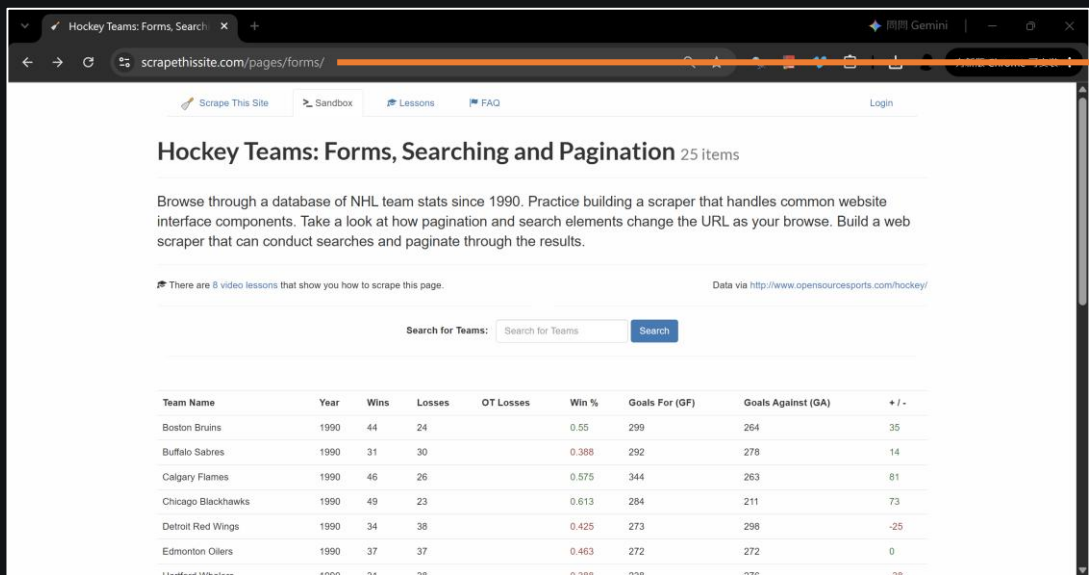
Request (爬蟲送出)

- 方法 Method : GET / POST
 - GET 取得資料
 - POST 提交資料，例如：帳號、密碼等 (包成 body)
- 網址 URL : 目標網站
- 標頭 Headers : 說明來意

Response (伺服器回來)

- 狀態碼 Status : 是否順利
- 標頭 Headers : 內容型別、編碼
- 本文 Body : HTML / JSON (.text)

URL 結構



https://www.scrapethissite.com/pages/forms/

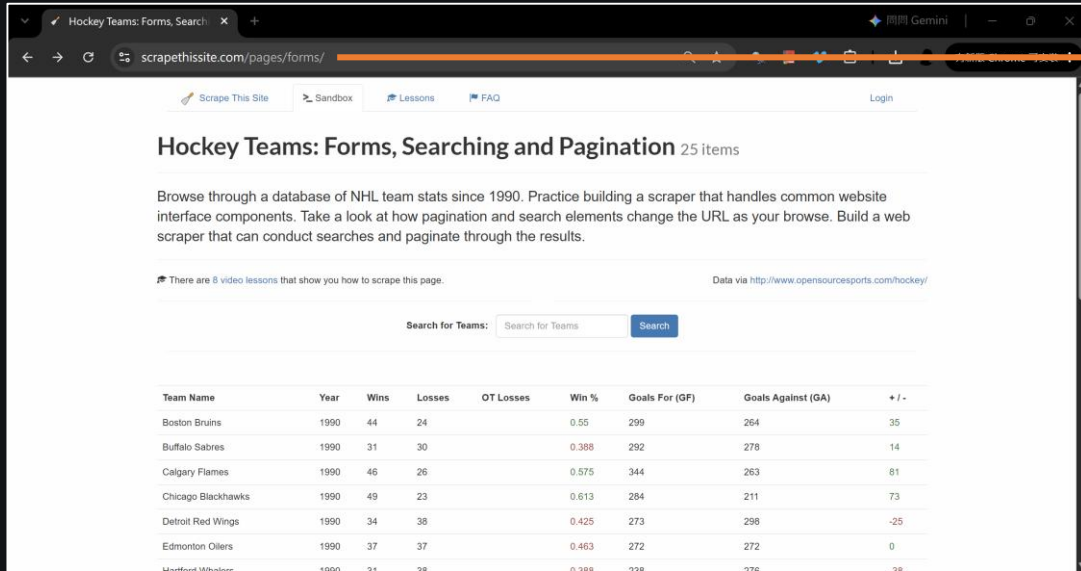
用戶端與網站伺服器之間傳輸網頁資料的**通訊協定**(Protocol/Scheme)

- HyperText Transfer Protocol Secure 超文本傳輸安全協定
- HTTP : 資料傳輸明文沒有加密
HTTPS : 資料傳輸加上 SSL 或 TLS 憑證機制

衍生閱讀

- Amazon Web Services, [HTTP 與 HTTPS 之間有什麼區別？](#)

URL 結構



<https://www.scrapethissite.com/pages/forms/>

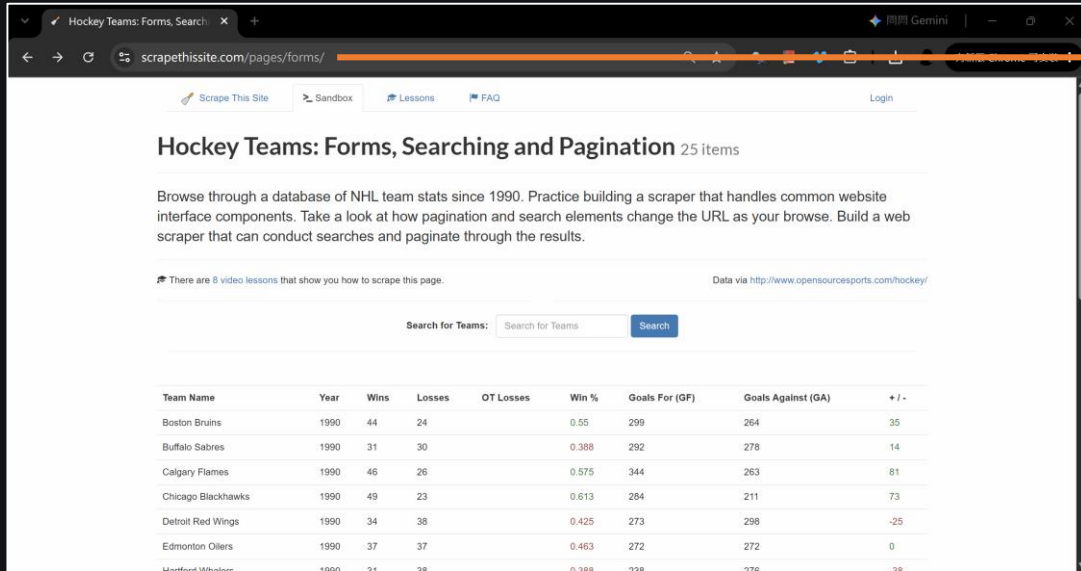
子網域(Subdomain) 與 網域名稱(Domain Name)

➤ www : World Wide Web

➤ 網域名稱包含網站主名稱與頂級域名

- ✓ 網站主名稱可自定，用來取代 IP 位址(Internet Protocol Address)的數字，通常是品牌或公司名稱。
- ✓ 頂級域名：.com 商業 / .gov 政府 / .edu 教育 / .org 非營利。
- ✓ 國碼(非必要)：.tw 台灣 / .jp 日本 / .uk 英國 /

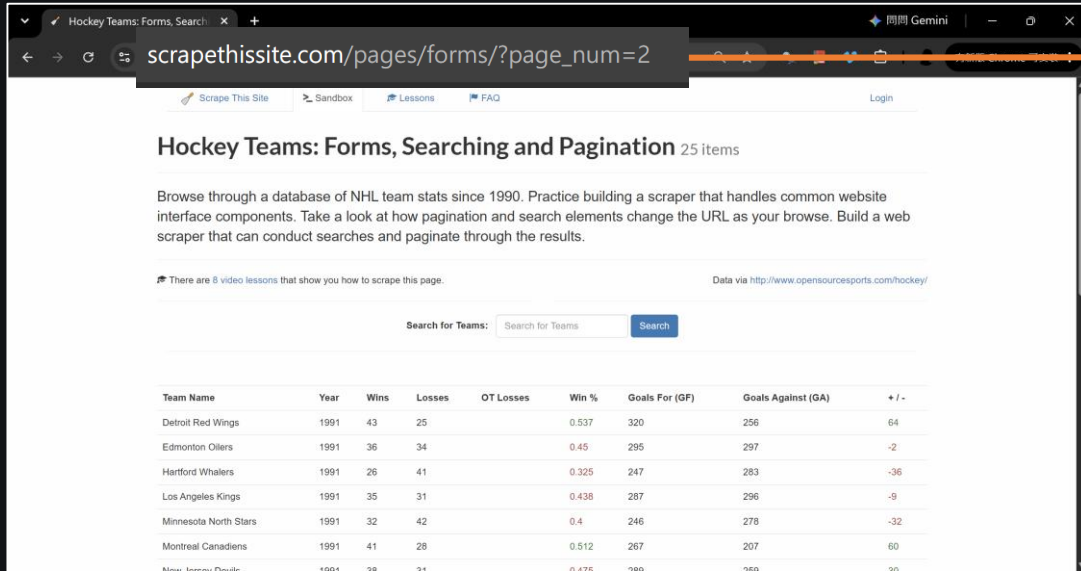
URL 結構



`https://www.scrapethissite.com/pages/forms/`

路徑(Path) 為指向網站內部特定網頁或檔案的階層位置

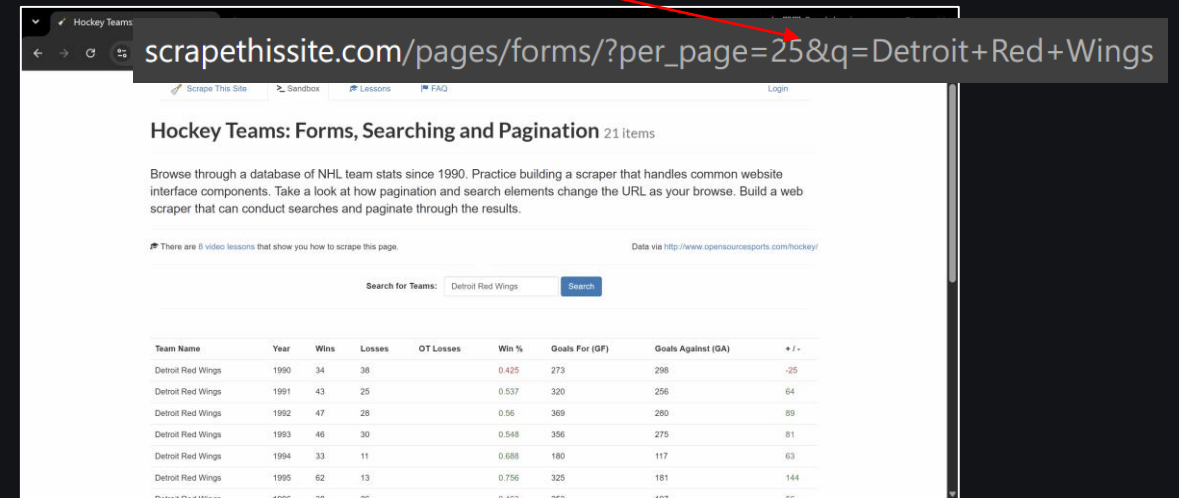
URL 結構



`https://www.scrapethissite.com/pages/forms/?page_num=2`

參數 (Parameters) 用於表示指定頁面特定跳轉位置的附加符號

➤ 以 `?` 開頭，且用 `&` 分隔的鍵值對。



URL 結構

YenLinWu/Python_Web_Crawling_Practical_Applications#課程大綱與教材

課程大綱與教材

週次	日期	主題	工具/技術	教材
W1	09/15	網路爬蟲概論與開發環境建置	VS Code · Python 3	Week 01
W2	09/22	Python 網路爬蟲程式設計基礎	虛擬環境 · pd.read_html 爬取資料表	Week 02
W3	09/29	HTTP 通訊協定與網頁資料請求	HTTP · requests	尚未發布
W4	10/06	HTML 網頁結構解析與資料擷取	BeautifulSoup · read_html	尚未發布
W5	10/13	正規表達式與文字資料擷取	正規表達式 re	尚未發布
W6	10/20	分頁爬取、資料儲存與爬蟲規範	分頁 · SQLite · robots.txt	尚未發布
W7	10/27	AI 輔助程式開發與產出驗證	Codeium/Windsurf · AI 輔助編程	尚未發布
W8	11/03	動態網頁爬取與 API 資料擷取	Selenium/Playwright · JSON · XML	尚未發布

https://github.com/YenLinWu/Python_Web_Crawling_Practical_Applications#課程大綱與教材

片段識別符 (Fragment / Hash) 用來控制網頁畫面的捲動位置或前端切換

- 以 **#** 來定位網頁內的特定區塊位置。
- 僅在用戶端的瀏覽器上運作，**#** 後面的內容部會傳給伺服器。

STATUS_CODE 狀態碼

類別	意思	爬蟲常見	誰的問題
2xx	成功	200 OK	沒問題
3xx	轉址	301 / 302 (requests 會自動跟過去)	網址改變
4xx	用戶端錯誤	403 拒絕、404 找不到、429 太頻繁	我們 (網址、參數、頻率)
5xx	伺服器錯誤	500 內部錯誤、503 暫停服務	對方 (稍後再試)

標頭 Headers

請求標頭 (我們送出)

- User-Agent : 標明是程式、用途
- Accept-Language : 採用語言
- Referer : 從哪一頁點過來
- Cookie : 保持登入狀態、繞過限制、取得權限等

回應標頭 (伺服器回來)

- Content-Type : text/html; charset=utf-8
→ 內容型別 + 編碼
- Content-Length : 內容大小
- Set-Cookie : 要瀏覽器記住的東西

衍生閱讀

- MDN Web Docs, [User-Agent 標頭](#)

Response 回應解頗

語法	說明 (範例值)
resp.status_code	狀態碼 (200)
resp.reason	狀態說明 (OK)
resp.headers.get("Content-Type")	內容型別與編碼 (text/html; charset=utf-8)
resp.encoding	.text 解碼用的編碼
resp.text	回應字串內容(HTML 原始碼)
resp.elapsed.total_seconds()	從送出到收到回應的秒數
resp.request.headers["User-Agent"]	我們送出的 User-Agent

```
import requests
```

```
URL = "https://www.scrapethissite.com/pages/forms/"  
HEADERS = {"User-Agent": "....."}
```

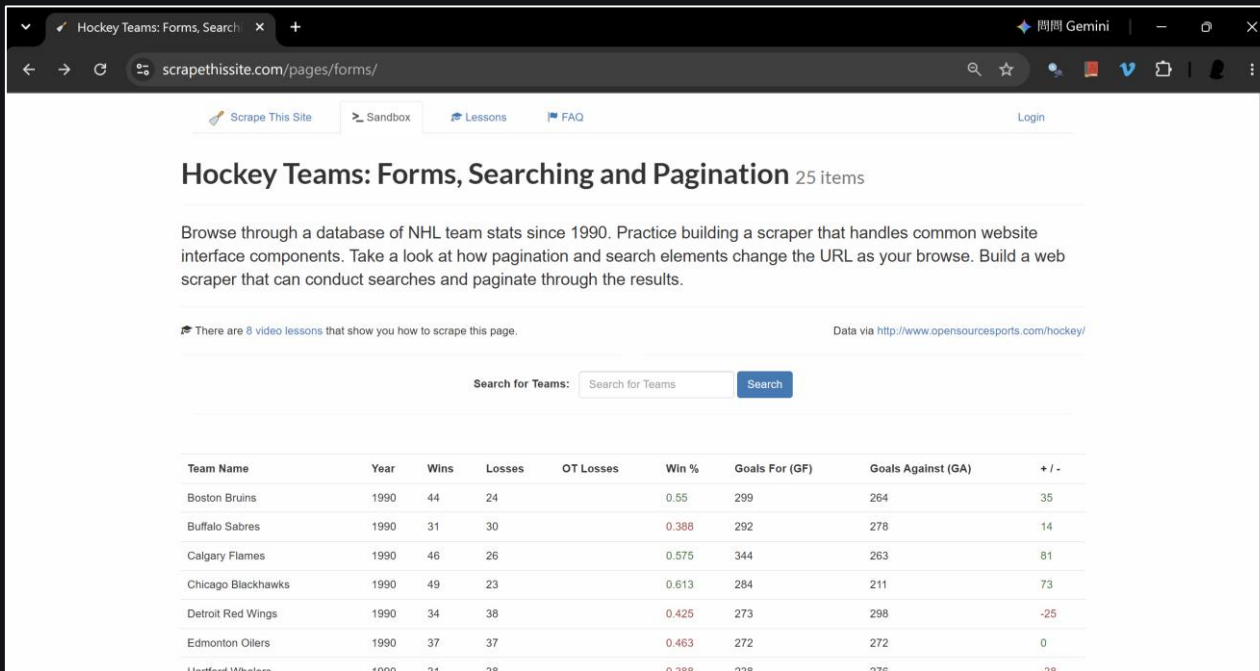
```
resp = requests.get(URL, headers=HEADERS, timeout=10)
```

```
.....
```

衍生閱讀

- Requests, [Quickstart: Response Content](#)

課堂練習 - 1



The screenshot shows a web browser window with the URL <https://www.scrapethissite.com/pages/forms/>. The page title is "Hockey Teams: Forms, Searching and Pagination 25 items". Below the title, there is a description: "Browse through a database of NHL team stats since 1990. Practice building a scraper that handles common website interface components. Take a look at how pagination and search elements change the URL as your browse. Build a web scraper that can conduct searches and paginate through the results." There is also a search bar with the text "Search for Teams:" and a "Search" button. Below the search bar, there is a table of NHL team stats.

Team Name	Year	Wins	Losses	OT Losses	Win %	Goals For (GF)	Goals Against (GA)	+ / -
Boston Bruins	1990	44	24		0.55	299	264	35
Buffalo Sabres	1990	31	30		0.388	292	278	14
Calgary Flames	1990	46	26		0.575	344	263	81
Chicago Blackhawks	1990	49	23		0.613	284	211	73
Detroit Red Wings	1990	34	38		0.425	273	298	-25
Edmonton Oilers	1990	37	37		0.463	272	272	0
Hartford Whalers	1990	31	38		0.388	278	276	2

<https://www.scrapethissite.com/pages/forms/>

- 送出 Request 給 ScrapeThisSite 練習網站的冰球隊伍統計表，解析伺服器回應的 Response：

```
import requests

URL = "....."
HEADERS = {"User-Agent": "....."}

resp = requests.get(URL, headers=HEADERS, timeout=10)

# ----- Response ( 伺服器回應內容 ) -----
print(f"[Response] 狀態碼={resp.status_code} {resp.reason}")
print(f"[Response] Content-Type={resp.headers.get('Content-Type')}")
print(f"[Response] 編碼={resp.encoding}")
print(f"[Response] 花費秒數={resp.elapsed.total_seconds():.2f}")
print(f"[Response] User-Agent={resp.request.headers['User-Agent']}")
```

執行有誤時的例外處理

狀態碼	該做的事
403	確認 robots.txt / 使用條款
404	檢查網址與參數有沒有拼錯
5xx	等一下再試，不要連續狂送需求

衍生閱讀

- Requests, [Errors and Exceptions](#)

```
import requests

URL = "https://www.scrapethissite.com/pages/forms/"
HEADERS = {"User-Agent": "....."}

try:
    resp = requests.get(url, headers=HEADERS, timeout=10)
    resp.raise_for_status() # 4xx/5xx → HTTPError

except requests.exceptions.HTTPError as e:
    print("狀態碼", e.response.status_code)

except requests.exceptions.Timeout:
    print("等太久沒回應")

except requests.exceptions.ConnectionError:
    print("連不上：網路、DNS、防火牆/代理")

except requests.exceptions.TooManyRedirects:
    print("重新導向次數超過上限 ( 預設 30 次 )")

except requests.exceptions.RequestException as e:
    print("其他請求錯誤", e)
```

爬蟲程式的執行日誌 Log

需求	print	logging
事件何時發生？	無法得知	自動加時間戳記
是資訊還是錯誤？	分不出來	DEBUG / INFO / WARNING / ERROR / CRITICAL
程式關閉後是否可追查？	無法	寫進 .log 檔，事後可查
出錯時的完整原因	要自己印	logger.exception() 自動附 traceback
部署上線後只印重點	逐行刪除或註解 print	調整等級

Log 重要性：排查非預期錯誤、監控執行狀態、方便環境部署

Log 分級紀錄

等級	程度
INFO	正常流程紀錄，例如：啟動時間、結束時間、統計數據等。
WARNING	非致命警訊，例如：429 觸發限流，程式暫停後再試。
ERROR	可預期可處理的錯誤，例如：網頁中特定資訊無法爬取，跳過爬取其他資訊。
CRITICAL	整個程式癱瘓，例如：資料庫連線失敗、設定檔失效、本地 IP 遭封鎖等。

logging 模組

定義日誌排版格式

- %(asctime)s 代表執行時間
- [% (levelname)s] 是日誌等級
- %(name)s 是日誌器名稱
- %(message)s 訊息內容

```
import logging

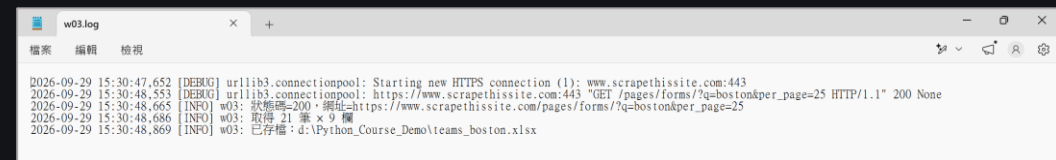
FMT = "%(asctime)s [% (levelname)s] %(name)s: %(message)s"
LOG_FILE = os.path.join(os.path.dirname(os.path.abspath(__file__)), "w03.log")

logging.basicConfig(
    level=logging.DEBUG,
    format=FMT,
    handlers=[
        logging.FileHandler(LOG_FILE, encoding="utf-8"),
        logging.StreamHandler(), # 同時印在螢幕
    ],
)

logger = logging.getLogger("w03")

logger.info("狀態碼=%s", resp.status_code)
```

Log 內容



```
w03.log
2026-09-29 15:30:47,652 [DEBUG] urllib3.connectionpool: Starting new HTTPS connection (1): www.scrapethissite.com:443
2026-09-29 15:30:48,553 [DEBUG] urllib3.connectionpool: https://www.scrapethissite.com:443 "GET /pages/forms/?q=boston&per_page=25 HTTP/1.1" 200 None
2026-09-29 15:30:48,665 [INFO] w03: 狀態碼=200 + 網址=https://www.scrapethissite.com/pages/forms/?q=boston&per_page=25
2026-09-29 15:30:48,686 [INFO] w03: 取得 21 筆 x 9 欄
2026-09-29 15:30:48,869 [INFO] w03: 已存檔: d:\Python_Course_Demo\teams_boston.xlsx
```

logging 模組

初始化全域設定

➤ level : 設定日誌的最低顯示門檻

若設定為 DEBUG，代表「所有等級」的日誌 (DEBUG、INFO、WARNING、ERROR、CRITICAL) 全部都要記錄，最為詳細。

➤ handlers : 設定輸出目的地

- ◆ FileHandler(LOG_FILE, encoding="utf-8") :
將 Log 寫入實體檔案，且指定 utf-8 編碼，防止中文產生亂碼。
- ◆ .log 檔案為純文字檔，可用文字編輯器開啟，例如：記事本、Notepad++ 等。
- ◆ StreamHandler() :
將 Log 同步印在螢幕(終端機)上，方便我們邊跑邊看。

```
import logging

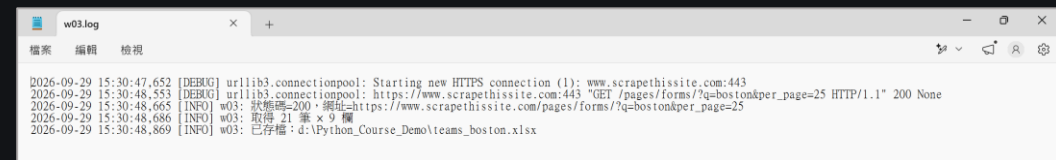
FMT = "%(asctime)s [%(levelname)s] %(name)s: %(message)s"
LOG_FILE = os.path.join(os.path.dirname(os.path.abspath(__file__)), "w03.log")

logging.basicConfig(
    level=logging.DEBUG,
    format=FMT,
    handlers=[
        logging.FileHandler(LOG_FILE, encoding="utf-8"),
        logging.StreamHandler(), # 同時印在螢幕
    ],
)

logger = logging.getLogger("w03")

logger.info("狀態碼=%s", resp.status_code)
```

Log 內容



```
2026-09-29 15:30:47,652 [DEBUG] urllib3.connectionpool: Starting new HTTPS connection (1): www.scrapethissite.com:443
2026-09-29 15:30:48,553 [DEBUG] urllib3.connectionpool: https://www.scrapethissite.com:443 "GET /pages/forms/?q=boston&per_page=25 HTTP/1.1" 200 None
2026-09-29 15:30:48,665 [INFO] w03: 狀態碼=200，網址=https://www.scrapethissite.com/pages/forms/?q=boston&per_page=25
2026-09-29 15:30:48,686 [INFO] w03: 取得 21 筆 x 9 欄
2026-09-29 15:30:48,869 [INFO] w03: 已存檔：d:\Python_Course_Demo\teams_boston.xlsx
```

logging 模組

建立專屬日誌紀錄器

- 往後使用 logger 輸出時，Log 便會自動標註 w03，方便在大型專案中識別是哪一個模組發出的訊息。

```
import logging

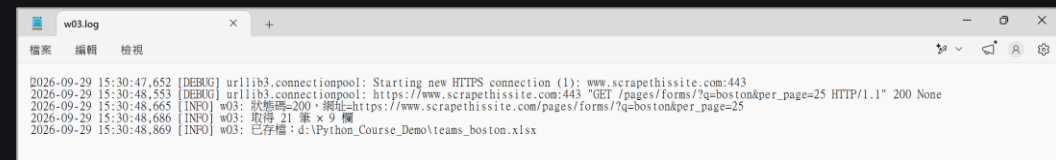
FMT = "%(asctime)s [%(levelname)s] %(name)s: %(message)s"
LOG_FILE = os.path.join(os.path.dirname(os.path.abspath(__file__)), "w03.log")

logging.basicConfig(
    level=logging.DEBUG,
    format=FMT,
    handlers=[
        logging.FileHandler(LOG_FILE, encoding="utf-8"),
        logging.StreamHandler(), # 同時印在螢幕
    ],
)

logger = logging.getLogger("w03")

logger.info("狀態碼=%s", resp.status_code)
```

Log 內容



```
w03.log
2026-09-29 15:30:47,652 [DEBUG] urllib3.connectionpool: Starting new HTTPS connection (1): www.scrapethissite.com:443
2026-09-29 15:30:48,553 [DEBUG] urllib3.connectionpool: https://www.scrapethissite.com:443 "GET /pages/forms/?q=boston&per_page=25 HTTP/1.1" 200 None
2026-09-29 15:30:48,665 [INFO] w03: 狀態碼=200，網址=https://www.scrapethissite.com/pages/forms/?q=boston&per_page=25
2026-09-29 15:30:48,686 [INFO] w03: 取得 21 筆 x 9 欄
2026-09-29 15:30:48,869 [INFO] w03: 已存檔：d:\Python_Course_Demo\teams_boston.xlsx
```

logging 模組

印出 Log 訊息

➤ 以 INFO (一般資訊) 等級輸出訊息。

```
import logging

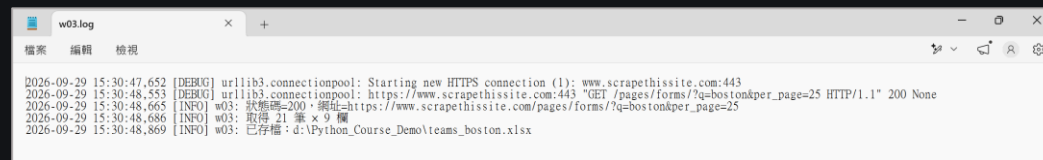
FMT = "%(asctime)s [%(levelname)s] %(name)s: %(message)s"
LOG_FILE = os.path.join(os.path.dirname(os.path.abspath(__file__)), "w03.log")

logging.basicConfig(
    level=logging.DEBUG,
    format=FMT,
    handlers=[
        logging.FileHandler(LOG_FILE, encoding="utf-8"),
        logging.StreamHandler(), # 同時印在螢幕
    ],
)

logger = logging.getLogger("w03")

logger.info("狀態碼=%s", resp.status_code)
```

Log 內容



```
w03.log
2026-09-29 15:30:47,652 [DEBUG] urllib3.connectionpool: Starting new HTTPS connection (1): www.scrapethissite.com:443
2026-09-29 15:30:48,553 [DEBUG] scrapy.core.spidermiddlewares.downloader: GET https://www.scrapethissite.com:443 "GET /pages/forms/?q=boston&per_page=25 HTTP/1.1" 200 None
2026-09-29 15:30:48,665 [INFO] w03: 狀態碼=200，網址=https://www.scrapethissite.com/pages/forms/?q=boston&per_page=25
2026-09-29 15:30:48,686 [INFO] w03: 取得 21 筆 x 9 欄
2026-09-29 15:30:48,869 [INFO] w03: 已存檔：d:\Python_Course_Demo\teams_boston.xlsx
```

logging 模組

印出 Log 訊息

➤ 以 INFO (一般資訊) 等級輸出訊息。

```
import logging

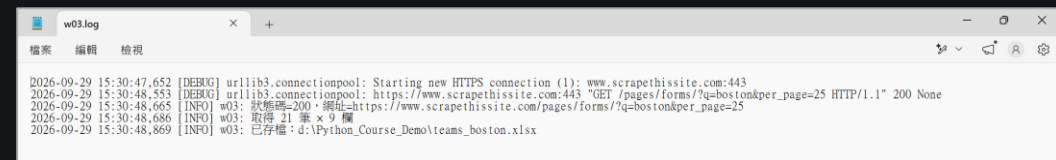
FMT = "%(asctime)s [%(levelname)s] %(name)s: %(message)s"
LOG_FILE = os.path.join(os.path.dirname(os.path.abspath(__file__)), "w03.log")

logging.basicConfig(
    level=logging.DEBUG,
    format=FMT,
    handlers=[
        logging.FileHandler(LOG_FILE, encoding="utf-8"),
        logging.StreamHandler(), # 同時印在螢幕
    ],
)

logger = logging.getLogger("w03")

logger.info("狀態碼=%s", resp.status_code)
```

Log 內容



```
w03.log
2026-09-29 15:30:47,652 [DEBUG] urllib3.connectionpool: Starting new HTTPS connection (1): www.scrapethissite.com:443
2026-09-29 15:30:48,553 [DEBUG] scrapy.core.spidermiddlewares.downloader: GET https://www.scrapethissite.com:443 "GET /pages/forms/?q=boston&per_page=25 HTTP/1.1" 200 None
2026-09-29 15:30:48,665 [INFO] w03: 狀態碼=200 + 網址=https://www.scrapethissite.com/pages/forms/?q=boston&per_page=25
2026-09-29 15:30:48,686 [INFO] w03: 取得 21 筆 x 9 欄
2026-09-29 15:30:48,869 [INFO] w03: 已存檔: d:\Python_Course_Demo\teams_boston.xlsx
```

try / except 結合 logging

```
import os
import logging
import requests

FMT = "%(asctime)s [%(levelname)s] %(name)s: %(message)s"
LOG_FILE = os.path.join(os.path.dirname(os.path.abspath(__file__)), "w03.log")

logging.basicConfig(
    level=logging.DEBUG,
    format=FMT,
    handlers=[
        logging.FileHandler(LOG_FILE, encoding="utf-8"),
        logging.StreamHandler(),
    ],
)

logger = logging.getLogger("w03")

URL = "...."
HEADERS = {"User-Agent": "....."}
```

```
try:
    resp = requests.get(URL, headers=HEADERS, timeout=10)
    resp.raise_for_status()
    logger.info("狀態碼=%s", resp.status_code)

except requests.exceptions.HTTPError as e:
    logger.error("狀態碼=%s", e.response.status_code)

except requests.exceptions.Timeout:
    logger.error("等太久沒回應")

except requests.exceptions.ConnectionError:
    logger.error("連不上：網路、DNS、防火牆/代理")

except requests.exceptions.TooManyRedirects:
    logger.error("重新導向次數超過上限 ( 預設 30 次 ) ")

except requests.exceptions.RequestException:
    logger.exception("其他請求錯誤")
```

課堂練習 - 2

- 爬取 ScrapeThisSite 練習網站的冰球隊伍統計表
- 將表格資料存成 output.csv 檔

Hockey Teams: Forms, Searching and Pagination 25 items

Browse through a database of NHL team stats since 1990. Practice building a scraper that handles common website interface components. Take a look at how pagination and search elements change the URL as you browse. Build a web scraper that can conduct searches and paginate through the results.

There are 8 video lessons that show you how to scrape this page. Data via <http://www.opensourcesports.com/hockey/>

Search for Teams: Search

Team Name	Year	Wins	Losses	OT Losses	Win %	Goals For (GF)	Goals Against (GA)	+ / -
Boston Bruins	1990	44	24		0.55	299	264	35
Buffalo Sabres	1990	31	30		0.388	292	278	14
Calgary Flames	1990	46	26		0.575	344	263	81
Chicago Blackhawks	1990	49	23		0.613	284	211	73
Detroit Red Wings	1990	34	38		0.425	273	298	-25
Edmonton Oilers	1990	37	37		0.463	272	272	0
Hartford Whalers	1990	31	38		0.388	238	276	-38
Los Angeles Kings	1990	46	24		0.575	340	254	86
Minnesota North Stars	1990	27	39		0.338	256	266	-10
Montreal Canadiens	1990	39	30		0.487	273	249	24
New Jersey Devils	1990	32	33		0.4	272	264	8
New York Islanders	1990	25	45		0.312	223	290	-67
New York Rangers	1990	36	31		0.45	297	265	32
Philadelphia Flyers	1990	33	37		0.412	252	267	-15
Pittsburgh Penguins	1990	41	33		0.512	342	305	37
Quebec Nordiques	1990	16	50		0.2	236	354	-118
St. Louis Blues	1990	47	22		0.588	310	250	60
Toronto Maple Leafs	1990	23	46		0.287	241	318	-77
Vancouver Canucks	1990	28	43		0.35	243	315	-72
Washington Capitals	1990	37	36		0.463	258	258	0
Winnipeg Jets	1990	26	43		0.325	260	288	-28
Boston Bruins	1991	36	32		0.45	270	275	-5
Buffalo Sabres	1991	31	37		0.388	289	299	-10
Calgary Flames	1991	31	37		0.388	296	305	-9
Chicago Blackhawks	1991	36	29		0.45	257	236	21

<https://www.scrapethissite.com/pages/forms/>

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Team Name	Year	Wins	Losses	OT Losses	Win %	Goals For (GF)	Goals Against (GA)	+ / -										
2	Boston Bruins	1990	44	24		0.55	299	264	35										
3	Buffalo Sabres	1990	31	30		0.388	292	278	14										
4	Calgary Flames	1990	46	26		0.575	344	263	81										
5	Chicago Blackhawks	1990	49	23		0.613	284	211	73										
6	Detroit Red Wings	1990	34	38		0.425	273	298	-25										
7	Edmonton Oilers	1990	37	37		0.463	272	272	0										
8	Hartford Whalers	1990	31	38		0.388	238	276	-38										
9	Los Angeles Kings	1990	46	24		0.575	340	254	86										
10	Minnesota North Stars	1990	27	39		0.338	256	266	-10										
11	Montreal Canadiens	1990	39	30		0.487	273	249	24										
12	New Jersey Devils	1990	32	33		0.4	272	264	8										
13	New York Islanders	1990	25	45		0.312	223	290	-67										
14	New York Rangers	1990	36	31		0.45	297	265	32										
15	Philadelphia Flyers	1990	33	37		0.412	252	267	-15										
16	Pittsburgh Penguins	1990	41	33		0.512	342	305	37										
17	Quebec Nordiques	1990	16	50		0.2	236	354	-118										
18	St. Louis Blues	1990	47	22		0.588	310	250	60										
19	Toronto Maple Leafs	1990	23	46		0.287	241	318	-77										
20	Vancouver Canucks	1990	28	43		0.35	243	315	-72										
21	Washington Capitals	1990	37	36		0.463	258	258	0										
22	Winnipeg Jets	1990	26	43		0.325	260	288	-28										
23	Boston Bruins	1991	36	32		0.45	270	275	-5										
24	Buffalo Sabres	1991	31	37		0.388	289	299	-10										
25	Calgary Flames	1991	31	37		0.388	296	305	-9										
26	Chicago Blackhawks	1991	36	29		0.45	257	236	21										

- 將 log 日誌存成 w03.log 檔，且紀錄下面內容

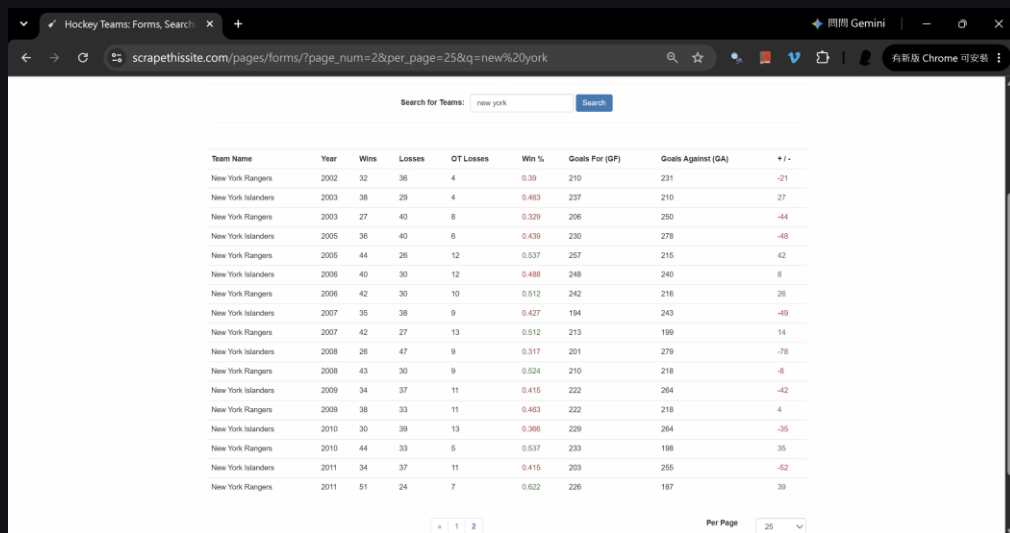
```
2026-09-29 15:30:47.652 [DEBUG] urllib3.connectionpool: Starting new HTTPS connection (1): www.scrapethissite.com:443
2026-09-29 15:30:48.553 [DEBUG] urllib3.connectionpool: https://www.scrapethissite.com:443 "GET /pages/forms/?q=boston&per_page=25 HTTP/1.1" 200 None
2026-09-29 15:30:48.665 [INFO] w03: 狀態碼=200，網址=https://www.scrapethissite.com/pages/forms/?q=boston&per_page=25
2026-09-29 15:30:48.686 [INFO] w03: 取得 21 筆 x 9 欄
2026-09-29 15:30:48.869 [INFO] w03: 已存檔：d:\Python_Course_Demo\teams_boston.xlsx
```

節 2

requests 帶參數

params 參數

params 參數就是「查詢條件」，同一個網址路徑，換不同參數就拿到不同資料，讓爬蟲程式變得可重複、可批次作業。



Team Name	Year	Wins	Losses	OT Losses	Win %	Goals For (GF)	Goals Against (GA)	+/-
New York Rangers	2002	32	36	4	0.39	231	231	-21
New York Islanders	2003	38	29	4	0.463	237	210	27
New York Rangers	2003	27	40	8	0.329	206	250	-44
New York Islanders	2005	36	40	6	0.439	230	278	-48
New York Rangers	2005	44	26	12	0.537	257	215	42
New York Islanders	2006	40	30	12	0.488	248	240	8
New York Rangers	2006	42	30	10	0.512	242	216	26
New York Islanders	2007	35	38	9	0.427	194	243	-49
New York Rangers	2007	42	27	13	0.512	213	199	14
New York Islanders	2008	26	47	9	0.317	201	279	-78
New York Rangers	2008	43	30	9	0.524	210	218	-8
New York Islanders	2009	34	37	11	0.415	222	264	-42
New York Rangers	2009	38	33	11	0.463	222	218	4
New York Islanders	2010	30	39	13	0.366	229	264	-35
New York Rangers	2010	44	33	5	0.537	233	198	35
New York Islanders	2011	34	37	11	0.415	203	255	-52
New York Rangers	2011	51	24	7	0.622	226	187	39

```
import requests
```

```
URL = "https://www.scrapethissite.com/pages/forms/"
```

```
HEADERS = {"User-Agent": "..."}  
PARAMS = {"q": "new york",
```

```
          "per_page": 25,  
          "page_num": 2}
```

```
resp = requests.get(URL, headers=HEADERS, params=PARAMS, timeout=20)
```

```
print(resp.url)      # ← 驗證實際送出的網址
```

params 負責組網址，可用 resp.url 來驗證。

衍生閱讀

- Requests, [Passing Parameters In URLs](#)

GET 與 POST：參數怎麼放

	GET + params=	POST + data=
參數位置	網址 ? 後面	本文 body
常見用途	搜尋、翻頁、篩選 (查資料)	登入、送出表單 (送資料)
可書籤/分享	可以	不行

課堂練習 - 3

- 爬取 ScrapeThisSite 練習網站的冰球隊伍統計表
- 只須爬取 Los Angeles Kings 與 Detroit Red Wings 這兩隊的資訊
- 將表格資料存成 output.xlsx 檔

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There are 8 video lessons that show you how to scrape this page. Data via <http://www.opensourcesports.com/hockey/>

Search for Teams: Search

Team Name	Year	Wins	Losses	OT Losses	Win %	Goals For (GF)	Goals Against (GA)	+ / -
Boston Bruins	1990	44	24		0.55	299	264	35
Buffalo Sabres	1990	31	30		0.388	292	278	14
Calgary Flames	1990	46	26		0.575	344	263	81
Chicago Blackhawks	1990	49	23		0.613	284	211	73
Detroit Red Wings	1990	34	38		0.425	273	298	-25
Edmonton Oilers	1990	37	37		0.463	272	272	0
Hartford Whalers	1990	31	38		0.388	278	276	2

<https://www.scrapethissite.com/pages/forms/>

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Team Name	Year	Wins	Losses	OT Losses	Win %	Goals For (GF)	Goals Against (GA)	+ / -										
2	Los Angeles Kings	1990	46	24		0.575	340	254	86										
3	Los Angeles Kings	1991	35	31		0.438	287	296	-9										
4	Los Angeles Kings	1992	39	35		0.464	338	340	-2										
5	Los Angeles Kings	1993	27	45		0.321	294	322	-28										
6	Los Angeles Kings	1994	16	23		0.333	142	174	-32										
7	Los Angeles Kings	1995	24	40		0.293	256	302	-46										
8	Los Angeles Kings	1996	28	43		0.341	214	268	-54										
9	Los Angeles Kings	1997	38	33		0.463	227	225	2										
10	Los Angeles Kings	1998	32	45		0.39	189	222	-33										
11	Los Angeles Kings	1999	39	27	4	0.476	245	228	17										
12	Los Angeles Kings	2000	38	28	3	0.463	252	228	24										
13	Los Angeles Kings	2001	40	27	4	0.488	214	190	24										
14	Los Angeles Kings	2002	33	37	6	0.402	203	221	-18										
15	Los Angeles Kings	2003	28	29	9	0.341	205	217	-12										
16	Los Angeles Kings	2005	42	35	5	0.512	249	270	-21										
17	Los Angeles Kings	2006	27	41	14	0.329	227	283	-56										
18	Los Angeles Kings	2007	32	43	7	0.39	231	266	-35										
19	Los Angeles Kings	2008	34	37	11	0.415	207	234	-27										
20	Los Angeles Kings	2009	46	27	9	0.561	241	219	22										
21	Los Angeles Kings	2010	46	30	6	0.561	219	198	21										
22	Los Angeles Kings	2011	40	27	15	0.488	194	179	15										
23	Detroit Red Wings	1990	34	38		0.425	273	298	-25										
24	Detroit Red Wings	1991	43	25		0.537	320	256	64										
25	Detroit Red Wings	1992	47	28		0.56	369	280	89										

節 3

開發者工具：Network 面板

Network 操作步驟

1

F12 開啟

切到「網路 /
Network」分頁

2

設定

勾 Preserve log ·
篩選 Doc

3

操作網頁

4

點開請求

Headers : Request
URL / Method /
Status

5

查詢參數

查看酬載

Network 操作步驟

1

F12 開啟

切到「網路 /
Network」分頁

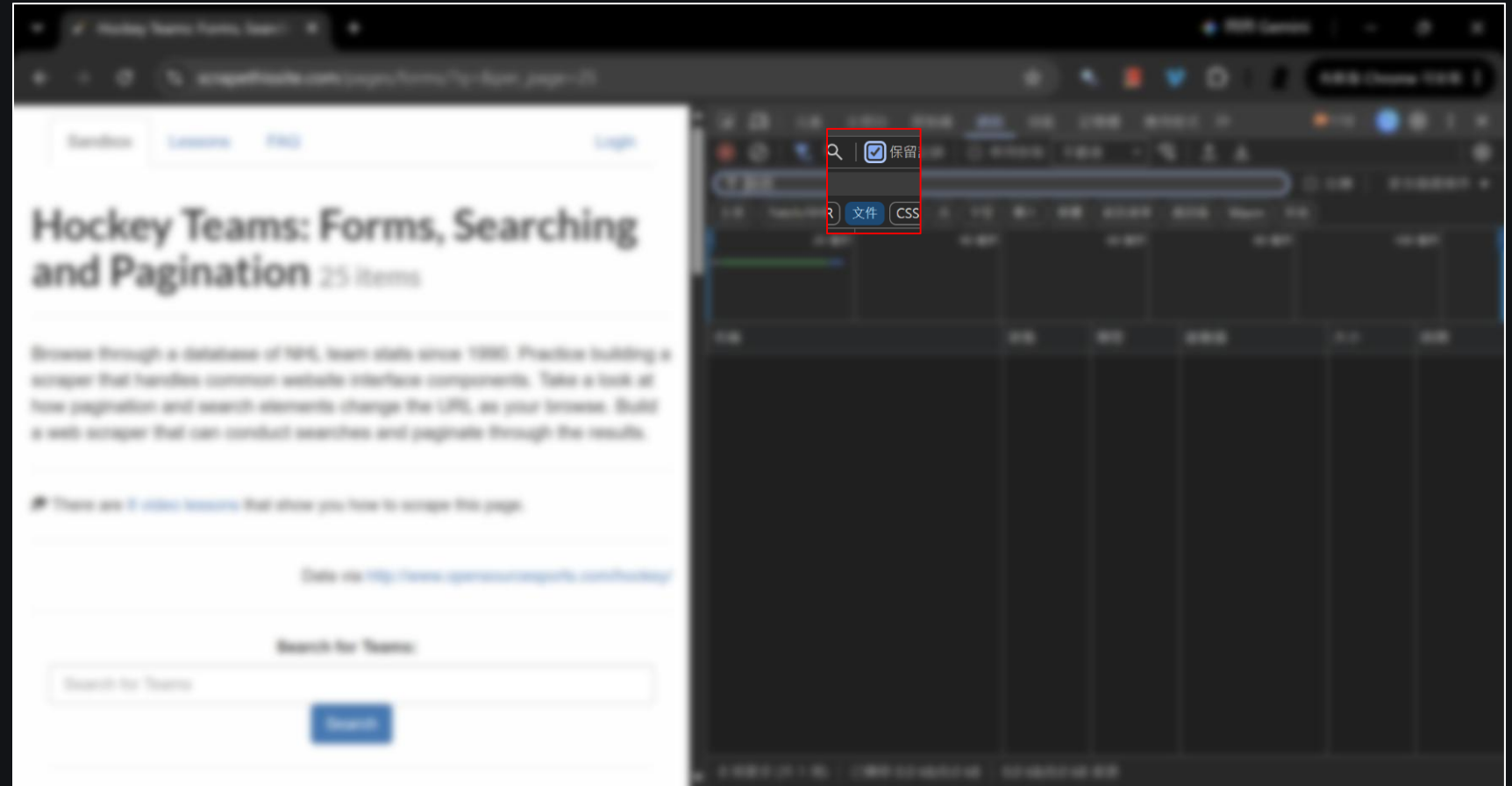
The screenshot shows a web browser window with the URL `scrapethissite.com/pages/forms/?q=&per_page=25`. The page title is "Hockey Teams: Forms, Searching and Pagination 25 items". The page content includes a description of the site, a search bar, and a "Search" button. The Chrome DevTools Network tab is open, showing a list of network requests. The first request is selected, and its details are visible in the right pane. The details pane shows the request type as "Fetch/XHR" and the status as "200 OK". The response is a JSON array of 25 items, which is displayed in the right pane. The status bar at the bottom of the DevTools panel shows "0 項要求 (共 1 項) | 已轉移 0.0 kB/0.0 kB | 0.0 kB/0.0 kB 資源".

Network 操作步驟

2

設定

勾 Preserve log
篩選 Doc



Network 操作步驟

3

操作網頁

The screenshot shows a web browser window with the URL `scrapethissite.com/pages/forms/?q=new+york&per_page=25`. The page content includes a title "Hockey Teams: Forms, Searching and Pagination 25 items", a description about building a scraper for NHL team stats, and a search form with the text "new york" entered. The search button is labeled "Search".

On the right side, the Chrome DevTools Network tab is open, displaying a list of network requests. The first three requests are highlighted:

名稱	狀態	類型	啟動器	大小	時間
tr/	200	document	其他	0.0 kB	84 毫秒
forms/?q=new+york&per_page=25	200	document	其他	6.0 kB	1.07 秒
tr/	200	document	forms/?q=new+york&	0.0 kB	22 毫秒

At the bottom of the Network tab, a summary bar shows: "3 項要求 (共 38 項) | 已轉移 6.0 kB/160 kB | 44.1 kB/2,432 kB 資源 | 完成: 1.59 秒 | DOMContentLoaded: 1.10 秒".

Network 操作步驟

4

點開請求

Headers : Request
URL / Method /
Status

The screenshot shows a web browser window with the URL `scrapethissite.com/pages/forms/?q=new+york&per_page=25`. The page content includes a title "Hockey Teams: Forms, Searching and Pagination 25 items", a description of the site's purpose, and a search form with the text "new york" entered. The browser's developer tools are open, displaying the network tab. The selected request is `forms/?q=new+york&per_page=25`. The request details show a GET method with a status of 200 OK. The response headers are visible, including `Alt-Svc`, `Cf-Cache-Status`, `Cf-Ray`, `Content-Encoding`, `Content-Type`, `Date`, and `Nel`.

名稱	標頭	預覽	回應	啟動器	時間	Cookie
tr/	▼ 一般					
forms/?q=new+york&per_page=25	要求網址		https://www.scrapethissite.com/pages/forms/?q=new+york&per_page=25			
tr/	要求方法		GET			
	狀態碼		200 OK			
	遠端位址		[2606:4700:3031::ac43:bb32]:443			
	參照網址政策		strict-origin-when-cross-origin			
	▼ 回應標頭					
	Alt-Svc		h3=":443"; ma=86400			
	Cf-Cache-Status		DYNAMIC			
	Cf-Ray		a45eae61e249c41-SIN			
	Content-Encoding		zstd			
	Content-Type		text/html; charset=utf-8			
	Date		Mon, 05 Oct 2026 19:02:53 GMT			
	Nel		{"report_to": "heroku-nel", "response_headers":			

Network 操作步驟

5

查詢參數

查看酬載

The screenshot shows a web browser window with the URL `scrapethissite.com/pages/forms/?q=new+york&per_page=25`. The page content includes a title "Hockey Teams: Forms, Searching and Pagination" followed by "25 items". Below the title, there is a description: "Browse through a database of NHL team stats since 1990. Practice building a scraper that handles common website interface components. Take a look at how pagination and".

The browser's developer tools are open, showing the Network tab. The selected request is `forms/?q=new+york&per_page=25`. The query parameters are visible:

名稱	值
q	new york
per_page	25

課堂練習 - 4

任務

1. 在冰球表頁搜尋 e
2. 每頁 100 筆，翻到第 3 頁
3. 在 Network 找出這次請求，填右表

欄位	我們看到的
Request URL	
Method	
Status Code	
參數	

<https://www.scrapethissite.com/pages/forms/>

今天，我們學到了什麼？

下週見 · Python 爬蟲實務應用